HOMESCOPE

Week 2 Feature Lists

PP4 Feature List and R&D Feature List

AI-Powered American House Price Dashboard

Feature planning document for a housing market dashboard that connects market filters, data visualization, model evaluation, and user-driven fair value prediction into a focused MVP scope.

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Course	Project & Portfolio IV
Assignment	Assignment 2 - Technical Research & Development Project
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Planning focus: separate final PP4 product scope from the smaller R&D demonstrator scope so the assignment stays focused on meaningful proof-of-technology features.

Project Overview

HomeScope is an AI-powered American house price dashboard designed to turn housing listings, market filters, data visualizations, and regression model results into simple market insight. The final product is built around three user goals: exploring market averages, comparing a listing against similar homes, and predicting a fair value estimate with model evaluation evidence.

PP4 Feature List - Final Project Scope

This list represents the planned final version of the Project & Portfolio IV project. Features are ordered by importance so the required MVP can be completed before stretch features are attempted.

1. Load the American Housing dataset from a local data folder or packaged dataset file.
2. Clean missing values in price, beds, baths, living space, city, state, county, population, density, income, latitude, and longitude fields.
3. Create model-ready engineered features such as price per square foot, bed-to-bath ratio, income-to-price ratio, and density bands.
4. Provide sidebar filters for state, city, minimum beds, minimum baths, and living space range.
5. Display a market snapshot with matching listings, average price, median price, and average price per square foot.
6. Display price distribution charts for the selected market filter.
7. Display price versus living space scatter charts to show relationship between home size and price.
8. Display city/state average price comparisons for the current filter.
9. Display national trend context using ASPUS average sales price data.
10. Train a baseline Linear Regression model for comparison.
11. Train a tree-based model such as Random Forest Regressor or Gradient Boosting Regressor.
12. Compare models using MAE, RMSE, and R-squared.
13. Display a residual plot for the best model to show where predictions succeed or fail.
14. Provide a prediction form for state, city, county, beds, baths, living space, and listing price.
15. Use city-level median demographics to keep the prediction form realistic while avoiding unnecessary user burden.
16. Display predicted fair value, listing price, and a simple market label such as below average, near average, above average, or possibly overpriced.
17. Display a prediction versus listing price chart after the user submits a sample property.
18. Organize code into reusable modules for data loading, data cleaning, filtering, model training, and prediction input creation.
19. Document final model strengths, limitations, and error patterns.
20. Prepare a clean demo experience for the final video walkthrough.

PP4 Stretch Features

- Interactive map using latitude and longitude.
- FHFA or FRED market index comparison as deeper trend context.
- Feature importance chart for the selected best model.
- Saved comparison examples using a local JSON or SQLite file.
- Deployment to Streamlit Community Cloud or a future React/FastAPI deployment path.
- Additional model comparison using Gradient Boosting or a small PyTorch neural network.

Features Intentionally Avoided for MVP

- Login screens and user authentication.

- Account settings and profile management.
- Dark mode as a feature requirement.
- Complex admin dashboards.
- Scraping Zillow, Redfin, Realtor.com, or other live listing platforms.
- Too many APIs before the local data and model pipeline is stable.

R&D Feature List - Assignment 2 Demonstrator

The R&D demonstrator only needs to prove the technology stack works interdependently. The app is intentionally smaller than the final PP4 product, but it still connects interface controls, local data, cleaning logic, visualizations, model training, model evaluation, and prediction output.

Category	R&D Feature
UI	Main Streamlit page with title, R&D caption, and four major sections: market snapshot, trend context, model evaluation, and prediction.
UI	Sidebar controls for state, city, minimum beds, minimum baths, minimum living space, and maximum living space.
UI	Metric cards for matching listings, average price, median price, and average price per square foot.
UI	Prediction form with state, city, county, beds, baths, living space, and listing price fields.
Data	Load American_Housing_Data_20231209.csv from the data folder.
Data	Load ASPUS.csv from the data folder for national trend context.
Data	Clean missing values, create price per square foot, remove unusable values, and prepare model-ready data.
Data	Filter market data based on user-selected sidebar controls.
Model	Train a baseline Linear Regression model.
Model	Train a Random Forest Regressor as the tree-based comparison model.
Model	Evaluate both models using MAE, RMSE, and R-squared.
Model	Select the best model by MAE and use it for the prediction form.
Visuals	Display price distribution, price versus living space, top average price by city, ASPUS trend line, residual plot, and prediction comparison chart.
Output	Display predicted fair value, listing price, and market label.
Code	Separate data utility and model utility logic from the Streamlit app file.
Scope	Exclude login, settings, and dark mode because they do not prove the unique HomeScope technology chain.

R&D Completion Evidence

- The app loads both local CSV files and presents data-driven outputs.
- Changing sidebar filters changes listing counts, summary metrics, and charts.
- Model training produces a visible comparison table for Linear Regression and Random Forest Regressor.
- The best model is selected by MAE and used in the prediction form.
- The prediction form produces a fair value estimate, a listing comparison, and a market label.
- The final note in the app explains that login, settings, and dark mode were excluded because they are not meaningful R&D interactions for HomeScope.